

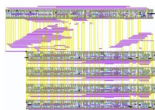
AI-Powered Analog Deep Learning Hardware Accelerator Design

Track: D: AI4AIchips

Zhiyang Ong¹ INSERT-YOUR-NAME!!!²

¹Department of Electrical and Computer Engineering, College of Engineering
Texas A&M University
College Station, TX
U.S.A.

²Department2, Division2
Organization2
City2, State2
Country2



June 8, 2026

Copyright © 2026 by Zhiyang Ong

This work is licensed under a
Creative Commons “Attribution-
NonCommercial-NoDerivatives 4.0
International” license.



Table of Contents

Front Matter

Acknowledgments

Project Information

Goal and Application

Design

Subsection 2.2

Section 3

Team Members' Experiences

Subsection 3.2

Acknowledgments

Francesco Stefanni, former labmate of Zhiyang Ong at University of Verona, who helped him craft $\text{\LaTeX}$ presentation slides.

The engineers of Mythic AI (Austin, TX) for showing us that analog in-memory computing is plausible, and their inspirational grit, from refusing to let their start-up go bankrupt multiple times, motivates us to be resilient in completing the IEEE SSCS Chipathon 2026 project.

Prof. Kaushik Sengupta at Princeton University and Prof. David Z Pan at The Awesome and Mighty University of Texas at Austin (trolling Aggieland haters, Go Longhorns!!!, Hook'em horns!!!) for inspiring us to craft AI-powered electronic design automation tools to kill IC design jobs.

Table of Contents

Front Matter

Acknowledgments

Project Information

Goal and Application

Design

Subsection 2.2

Section 3

Team Members' Experiences

Subsection 3.2

Project Information

Goal and Application

- ▶ Goals:
 - ▶ To develop a multi-modal AI-powered analog or radio frequency (RF) IC design flow
 - ▶ To apply the aforementioned AI-powered IC design flow for designing analog deep learning hardware accelerators
- ▶ Application: Analog deep learning hardware accelerators can be used for environmental monitoring, by training on analog signals captured from the sensors with data conversion between analog and digital signals.

Project Information

Design – High Level Proposal (2-3 sentences + Image)

Use semiconductor device models of memristors and floating-gate MOSFETS to approximate memristive and floating-gate MOSFET -based analog in-memory computing systems

Implement analog in-memory computing systems by replacing SRAM cell in SRAM subsystem with approximate RAM cell circuit model, based on the aforementioned semiconductor device models

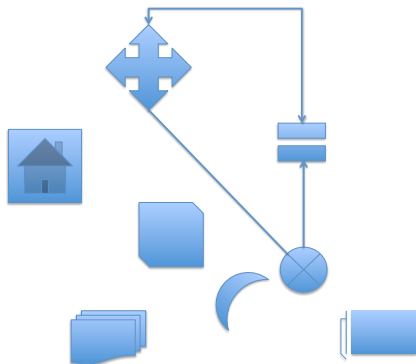


Figure: My caption [?, ?]

Slide Title 5

Slide Subtitle 5.

Statement 1 [?].

Statement 2.

Statement 3.

Table of Contents

Front Matter

Acknowledgments

Project Information

Goal and Application

Design

Subsection 2.2

Section 3

Team Members' Experiences

Subsection 3.2

Team Members' Experiences

- ▶ Zhiyang Ong:
 - ▶ 32-kbit synchronous SRAM with 32-bit words
 - ▶ MIPS processor architectures: single cycle, multi-cycle, 5-stage pipelined, 128-bit
 - ▶ Viterbi decoder, and encoder/decoder pairs for linear code
 - ▶ 64-bit Ladner-Fischer tree adder, using skew tolerant domino logic and logical effort
 - ▶ FIR filter from the bilinear transformation of an analog Butterworth filter
 - ▶ high frequency Colpitts oscillator, with an operating frequency of 20 MHz
 - ▶ audio amplifier: power & pre- amplifiers, active filters, & RC-filter bridge rectifier
 - ▶ EDA projects: gate sizing,, SRAM CAD for design space exploration, hardware generators for encoder/decoder pairs targeting linear code, logic simulator, PODEM ATPG

Slide Title 6

Slide Subtitle 6.

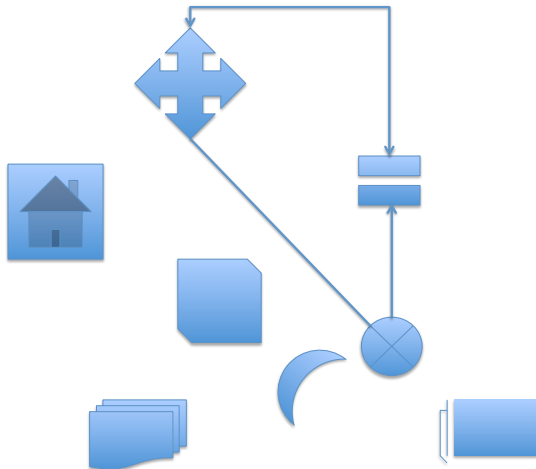


Figure: Caption #2

Questions, Suggestions, Doubts?

Statement 1 [?].

Statement 2.

Statement 3.

References